

Ivan Jimenez Pineda

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EDUCATION

Harvey Mudd College - Claremont, CA
Bachelor of Science in Engineering

| Expected Graduation: May 2028 |

PROFESSIONAL EXPERIENCE

CRPA Characterization Clinic Project - The Aerospace Corporation / Federal Aviation Administration | September 2026 - Present |

- Machined custom **mounting fixtures** for mechanical positioning turntables using manual operations, designing the hardware so the **Controlled Reception Pattern Antenna** may be placed securely inside the **anechoic chamber** for testing.
- Executed preliminary **benchtop RF test** and measurement procedures using a **Vector Network Analyzer** and dual probe antennas to prepare for upcoming element-level and array-level characterization of the CRPA.
- Collaborated with technical liaisons from The Aerospace Corporation and FAA to evaluate key parameters of a GPS CRPA anti-jamming antenna, supporting ongoing research to inform use of CRPAs for commercial aviation safety.

Mechanical Engineering Intern - ALLFAST Fastening Systems (Penn Aero) | May 2026 - August 2026 |

- Created **65 tooling layout drawings** and modeled parts for tooling assemblies using **SolidWorks** to support over **224 million parts** from the production backlog.
- Updated **BOMs** in the **ERP** system with accurate tooling numbers and quantities, enabling proper tracking and forecasting to mitigate heading downtime as a result of tooling shortages to allow for tools to be ordered quickly through **MRP**.
- Validated tooling layouts and BOM updates by physically assembling tooling and conducting reviews with machine operators and engineering teams to ensure accuracy.

Machine Shop Proctor / Workshop Proctor - Harvey Mudd College | May 2025 - Present |

- Instruct students on the safe operation of machining equipment (**Manual Mill, Lathe, CNC Router**), emphasizing adherence to process specifications and **tolerances down to 0.001"**.
- Train students to interpret technical drawings and apply **GD&T** to **optimize production processes** and reduce fabrication errors.
- Guided students through manual machining and metalworking processes to safely fabricate custom projects, including machined pens, rings, spinning tops, and sheet metal flowers.

PROJECTS

Autonomous Underwater Vehicle | August 2025 - May 2026 |

- Engineered and fabricated a **low-cost** autonomous **electro-mechanical robot** featuring a custom **winch mechanism** to deploy a tethered sensor payload for depth-resolved environmental data collection.
- Designed analog signal conditioning circuitry, utilizing unity gain and inverting offset operational amplifiers, to interface an **NTC thermistor, pressure sensor, and amorphous silicon solar cell** with a **Teensy microcontroller**.
- Programmed the Teensy using **Arduino IDE** for synchronized sensor logging and utilized **MATLAB** to calibrate raw data, capturing temperature and luminosity fluctuations relative to depth during coastal field deployments at Dana Point Harbor.

Brachistochrone Curve Model | February 2025 - May 2025 |

- Modeled cycloid, linear, and parabolic descent paths in **SolidWorks** using the **Equation Driven Curve** feature to optimize geometric accuracy and demonstrate kinematic principles for a client-requested physics demonstration.
- Fabricated the primary ramp structures and custom corner fixtures using a **CNC Router**, ensuring precise assembly and tolerances.

MITES Summer Scholar Symposium Project | June 2023 - July 2023 |

- Co-developed "Project Beatzthoven," an interactive **machine learning application** built in **Python** that algorithmically generates classical music utilizing Markov Chain logic.
- Applied **systematic debugging** practices and software engineering fundamentals to process data structures and balance competing project priorities within a strict 6-week development cycle to present at an MIT showcase.

The Machinist's Hammer | March 2025 - April 2025 |

- Machined a multi-material alloy steel hammer including a hard face, soft face, handle, and a stand (**AISI 4340 and AISI 1015**).
- Applied heat treatment (**857°C**) to hard face and verified results using a Rockwell hardness tester (**RC 43-48**).
- Used a mill, lathe, vertical bandsaw, belt sander, sand blaster, laser cutter, and 3D printer.

SKILLS

Programming: Python, C, SystemVerilog, LabVIEW, MATLAB, HTML, Arduino IDE, Data Acquisition (MyDAQ), VNA Analysis
Oscilloscope Testing, Git, Github, FPGA Design, CSS

Manufacturing: Precision Machining (Mill and Lathe), CNC Router, Laser Cutter, Waterjet, Heat Treatment, Rockwell Hardness
Testing, Sand Blasting, GD&T, SolidWorks, 3D Printing, FEA, Fusion 360 CAM, 3D Printing, Soldering

Languages: Fluent in English and Spanish